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Candidate-Accessible Evidence Portal with Per-Signal Explanation Pipeline for Remote Examination Integrity Findings

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Abstract

This disclosure describes a method and system for providing candidates of remotely-administered examinations with structured, candidate-accessible explanations of automated integrity findings generated against their examination sessions. Conventional remote-proctoring systems provide candidates with no visibility into the specific signals that triggered review, the confidence values associated with those signals, the underlying detector chains, or the contextual evidence available to human reviewers. This opacity is the documented primary driver of candidate distrust in remote-proctoring systems and creates significant procedural risk under data-protection regulations that require meaningful information about automated decision logic, under personnel-certification standards that require disclosure of automated tool use, and under emerging artificial-intelligence regulations that require transparency to subjects of automated decisions. This disclosure describes an architecture in which a per-signal explanation generator produces, for every monitoring signal contributing to a session risk score, a structured explanation record containing the event class, detector confidence, contributing signal-chain, and temporal context; an evidence-clip extractor produces synchronized audio-video-screen excerpts; and a candidate-accessible portal presents these explanations and clips to the candidate before any adverse certification decision is finalized, alongside a structured intake form by which the candidate may submit contextualization, dispute, or supplemental accommodation declaration. The candidate's responses are routed to the human reviewer alongside the original evidence packet.

1. Background

Remote-proctoring systems used in higher education, professional certification, and government testing routinely generate automated flags against candidate sessions. These flags are based on outputs of computer-vision, audio, behavioral, and statistical detectors. The flags may contribute to a session risk score, which in turn may trigger human review and may ultimately result in adverse action against the candidate — invalidation of the examination result, denial of certification, or referral for disciplinary process.

Existing systems are uniformly opaque to the candidate. The candidate is informed only that their session has been flagged for review, or in many cases is informed only after an adverse decision has already been taken. The candidate is not provided, before adverse action, with:

- the specific monitoring signals that contributed to the flag;
- the confidence values of those signals;
- the detector identities and signal chains;
- the time positions in the examination at which the signals occurred;
- the synchronized audio-video-screen evidence that the human reviewer is examining;
- the population-baseline or candidate-baseline against which the candidate's behavior is being compared;
- any opportunity to provide context, contextualization, or supplemental accommodation declaration relevant to the flagged behavior.

This opacity has been the documented primary driver of candidate distrust in remote-proctoring systems. It is incompatible with several emerging and current legal and regulatory regimes:

- **General Data Protection Regulation, Article 22**, which provides data subjects with the right to meaningful information about the logic of automated decision-making and the right to human intervention.
- **EU Artificial Intelligence Act, Articles 13 and 14**, which require transparency to subjects of high-risk AI systems and meaningful human oversight.
- **ISO/IEC 17024:2026** (personnel certification standard, AI provisions), which requires certification bodies to disclose AI use to candidates and to obtain candidate acknowledgment.
- Various jurisdiction-specific regulations on consumer-facing AI systems.

The technique disclosed herein addresses this regulatory and trust gap by providing the candidate with a structured, candidate-accessible portal containing the same per-signal evidence that human reviewers see, presented in a candidate-comprehensible format, before any adverse decision is finalized.

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2. Technique

2.1 Per-signal explanation generator

For each monitoring signal that contributes to a session risk score above a threshold of interest, a structured explanation record is generated. The explanation record contains:

(a) Event class identifier. A human-readable name for the event class — for example, "Multiple voices detected," "Phone visible in frame," "Sustained gaze away from screen," "Unusual answer-pattern statistical anomaly on item 28."

(b) Detector confidence. The detector's reported confidence in the event, normalized to a candidate-comprehensible scale (for example, "very high," "high," "moderate," "low") with the underlying numeric value also available.

(c) Signal chain. A description of how the event was produced: which detector, which model version, which thresholds were applied, what corroborating signals (if any) were considered.

(d) Temporal context. The exact time position in the examination at which the signal occurred, the duration of the signal, and the examination item (if any) that was being presented to the candidate at the time.

(e) Candidate-comprehensible rationale. A natural-language explanation of why this event class is considered relevant to examination integrity in general. For example: "Multiple distinct voices in the room during a closed-book examination may indicate that the candidate received assistance from another person."

(f) Alternative-explanation acknowledgment. A natural-language acknowledgment of common benign causes of the event, written in advance and stored in a per-event-class library. For example, for "Multiple voices detected": "This event may also occur if the candidate was reading aloud, if a family member or roommate entered the room, if a television or radio was audible in the background, if a pet vocalized, or if audio was leaking from another device. Please describe any such context below."

2.2 Evidence-clip extractor

For each explanation record, an evidence-clip extractor produces a synchronized excerpt of the original session data covering the temporal context of the event. The clip:

- has a configurable duration with default of 30 seconds (±15 seconds around the event);
- includes synchronized audio, video, and screen-capture data;
- is annotated with overlay markers indicating the precise moment of detector activation;
- is rendered in a candidate-accessible video format playable in standard web browsers;
- is served from access-controlled storage that records the candidate's viewing in the audit log.

2.3 Candidate-accessible portal

The candidate-accessible portal is a web-based interface authenticated to the candidate. Upon authentication, the candidate sees:

- a chronological timeline view of all explanation records associated with their session;
- for each explanation record, the candidate-comprehensible event description, the temporal context, the alternative-explanation acknowledgment, and a player for the evidence clip;
- a candidate-baseline comparison view, showing equivalent extracts from a baseline period of the same examination session in which no anomalies were detected, to support the candidate's understanding of how their behavior at the flagged moment differed from their baseline behavior;
- an aggregate session view, showing how the explanation records together contribute to the overall integrity finding.

The portal is made accessible to the candidate **before any adverse certification decision is finalized**. This is the critical distinguishing element from existing post-hoc disclosure regimes, in which a candidate is informed of the basis of an adverse decision only after the decision has been made and only in the context of an appeals process.

2.4 Structured intake form

For each explanation record, the candidate is provided with a structured intake form by which the candidate may submit:

- **Contextualization:** "Here is what was happening at this moment that may explain this event." Free-text field with optional supporting attachments.
- **Dispute:** "I dispute the accuracy of this detection." Structured options including "I was not doing what the system says I was doing," "The detection appears to be a technical artifact," "There is no second voice / object / behavior of the type detected."
- **Supplemental accommodation declaration:** "I have a relevant accommodation that was not declared in advance." Structured options linked to the accommodation taxonomy, with the candidate's acknowledgment that the certification body will review the declaration and request supporting documentation as appropriate.
- **No comment:** The candidate may decline to comment on any specific explanation record. Declining to comment must not be treated as an admission of misconduct.

2.5 Reviewer-facing routing

The candidate's responses to the structured intake form are routed to the human reviewer alongside the original evidence packet. The reviewer's interface presents:

- the original evidence packet (clip, signals, signal chains);
- the candidate's response to each explanation record (contextualization, dispute, supplemental accommodation declaration, or no comment);
- a structured adjudication form requiring the reviewer to record, for each explanation record, whether the candidate's response was considered, what weight was given to it, and what conclusion was reached.

2.6 Decision-finalization gate

The system enforces a procedural gate: the final integrity determination on the session may be recorded only after either (a) the candidate has interacted with the portal — viewing the explanation records and either submitting or declining to submit a response — or (b) a configurable waiting period has elapsed during which the candidate had documented opportunity to access the portal and did not do so. The waiting period is set in accordance with the applicable certification body's appeals-window policy.

2.7 Audit logging

Every candidate access to the portal, every viewing of an evidence clip, every submission of a structured intake response, and every reviewer adjudication is recorded in an immutable audit log. The audit log is retained in accordance with the applicable retention policy.

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3. Variant embodiments

The following variants are within the scope of the disclosed technique and are explicitly published as prior art:

(a) Application to live-proctored as well as record-and-review remote examinations.

(b) Application to in-person computerized examinations administered in test centers with automated monitoring.

(c) Integration with the candidate's existing certification-body account or with a federated identity provider.

(d) Multi-language rendering of the candidate-comprehensible event descriptions and alternative-explanation acknowledgments.

(e) Accessibility-compliant rendering of the portal in accordance with WCAG 2.2 AA or equivalent, including screen-reader support for textual explanations and accessible video player controls.

(f) Variants in which the portal is made accessible only to a subset of explanation records — for example, only those whose contribution to the session risk score exceeds a higher threshold — with the remaining records accessible only via formal appeal.

(g) Variants in which the candidate's response to the structured intake form is itself analyzed by a separate machine-learning classifier to detect duress, coercion, or templated responses; the output of such analysis must not be used as evidence of misconduct without separate human review.

(h) Variants in which the portal includes a candidate-initiated technical-difficulty escalation mechanism for cases in which the candidate believes the flagged event is the result of a system malfunction.

(i) Cross-jurisdictional variants that vary the portal's content and the candidate's available responses based on the candidate's jurisdiction of residence or examination administration, in accordance with local data-protection and disability law.

(j) Integration with the candidate's pre-examination accommodation declaration system, such that supplemental accommodation declarations submitted through the portal are routed for review under the same administrative process as pre-examination declarations.

(k) Integration with a fusion engine that combines behavioral telemetry with psychometric anomaly signals; in this variant, the explanation records include both behavioral and psychometric explanations on a per-examination-item basis.

(l) Privacy-preserving variants in which evidence clips are served via short-lived signed URLs and are not downloadable, to prevent unauthorized propagation.

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4. Operational considerations

4.1 Candidate identity verification

The candidate-accessible portal must verify the candidate's identity to the same standard as the candidate's original examination registration, to prevent unauthorized access to session evidence.

4.2 Reviewer guidance

Human reviewers must be trained to consider candidate responses in good faith, to weigh contextualization and accommodation declarations appropriately, and to avoid prejudicial conclusions from the absence of a response. The reviewer's adjudication form must require structured recording of how the candidate's response was considered.

4.3 Avoidance of evidence prejudice

The portal must not be used as an evidence-gathering tool against the candidate. Statements made by the candidate in the structured intake form must not be subject to interpretation as admissions of misconduct beyond what the candidate explicitly states. Candidates declining to respond must not be penalized.

4.4 Appeals-process integration

The candidate's interactions with the portal must be preserved as part of the appeals record. Subsequent formal appeals must include access to the same portal record. The portal does not replace the formal appeals process; it provides a pre-finalization opportunity for candidate input that supplements, rather than substitutes for, full appeals rights.

4.5 Retention and deletion

Evidence clips, explanation records, and candidate intake responses are subject to the same retention policy as the underlying session evidence. The candidate retains rights of access, rectification, and erasure under applicable data-protection law, subject to the certification body's legitimate retention requirements for audit and appeal purposes.

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5. Technical effects

The disclosed technique provides the following measurable improvements over conventional opaque remote-proctoring systems known in the art:

- Compliance with EU General Data Protection Regulation Article 22 right to meaningful information about automated-decision logic.
- Compliance with EU AI Act Articles 13 and 14 transparency and human-oversight requirements.
- Compliance with ISO/IEC 17024:2026 personnel-certification AI-disclosure provisions.
- Reduced appeal-cycle time, by surfacing legitimate candidate contextualization before adverse action rather than after.
- Reduced reviewer workload, by enabling reviewers to consider candidate context in their original adjudication rather than re-litigating the case during appeal.
- Improved candidate trust in the certification process, by replacing opaque automated decision-making with structured, transparent, candidate-accessible procedures.
- Reduced legal and reputational risk to certification bodies, by providing documented evidence that candidates were afforded procedural opportunity before adverse action.
- Improved adjudication quality, by routing the candidate's contemporaneous account to the reviewer alongside the original evidence, rather than separating them across an appeals timeline.

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6. Statement of public dedication

This disclosure is published expressly as prior art to prevent third parties from obtaining patent protection on the techniques described herein. The author dedicates the technical content of this disclosure to the public domain to the maximum extent permitted by law, and authorizes any party to practice the techniques described herein without restriction or royalty. The author claims no patent rights and will not seek any patent rights to the techniques disclosed.

End of defensive publication.